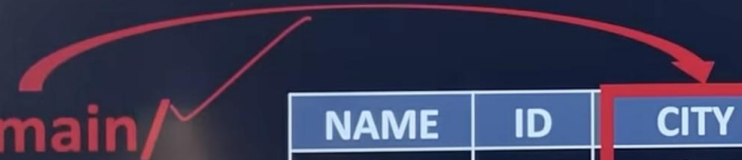


BASICS OF RDBMS

- **Domain (set of permissible value in particular column)** is a set of atomic values.
- By **atomic** we mean that each value in the domain is indivisible as far as the formal relational model is concerned.
- A common method of specifying a domain is to specify a data type from which the data values forming the domain are drawn.
- **E.g. Names:** The set of character strings that represent names of persons.

Domain/
Field/
Column/
Attribute



NAME	ID	CITY	COUNTRY	HOBBY
NISHA	1	AGRA	INDIA	PLAYING
NIKITA	2	DELHI	INDIA	DANCING
AJAY	3	AGRA	INDIA	CHESS
ARPIT	4	PATNA	INDIA	READING

Chapter-3 (Functional Dependency)

Basics & Properties

- Table (Relation) - A Relation is a set of tuples/rows/entities/records.

- Tuple - Each row of a Relation/Table is called Tuple.

- Arity/Degree - No. of columns/attributes of a Relation. E.g. - Arity is 5 in Table Student.

- Cardinality - No of rows/tuples/record of a Relational instance. E.g. - Cardinality is 4 in table Student.

Rows/Tuples/Record/
Cardinality →

NAME	ID	CITY	COUNTRY	HOBBY
NISHA	1	AGRA	INDIA	PLAYING
NIKITA	2	DELHI	INDIA	DANCING
AJAY	3	AGRA	INDIA	CHESS
ARPIT	4	PATNA	INDIA	READING

Properties of Relational tables

1. Cells contains atomic values
2. Values in a column are of the same kind
3. Each row is unique
4. No two tables can have the same name in a relational schema.
5. Each column has a unique name
6. The sequence of rows is insignificant
7. The sequence of columns is insignificant.



- Insertion anomalies: An independent piece of information cannot be recorded into a relation unless an irrelevant information must be inserted together at the same time.

Roll no	name	Age	Br_code	Br_name	Br_hod_name
1	A	19	101	Cs	Abc
2	B	18	101	Cs	Abc
3	C	20	101	Cs	Abc
4	D	20	102	Ec	Pqr

Chapter-3 (Functional Dependency)

Update Anomalies

- **Modification anomalies:** The update of a piece of information must occur at multiple locations.

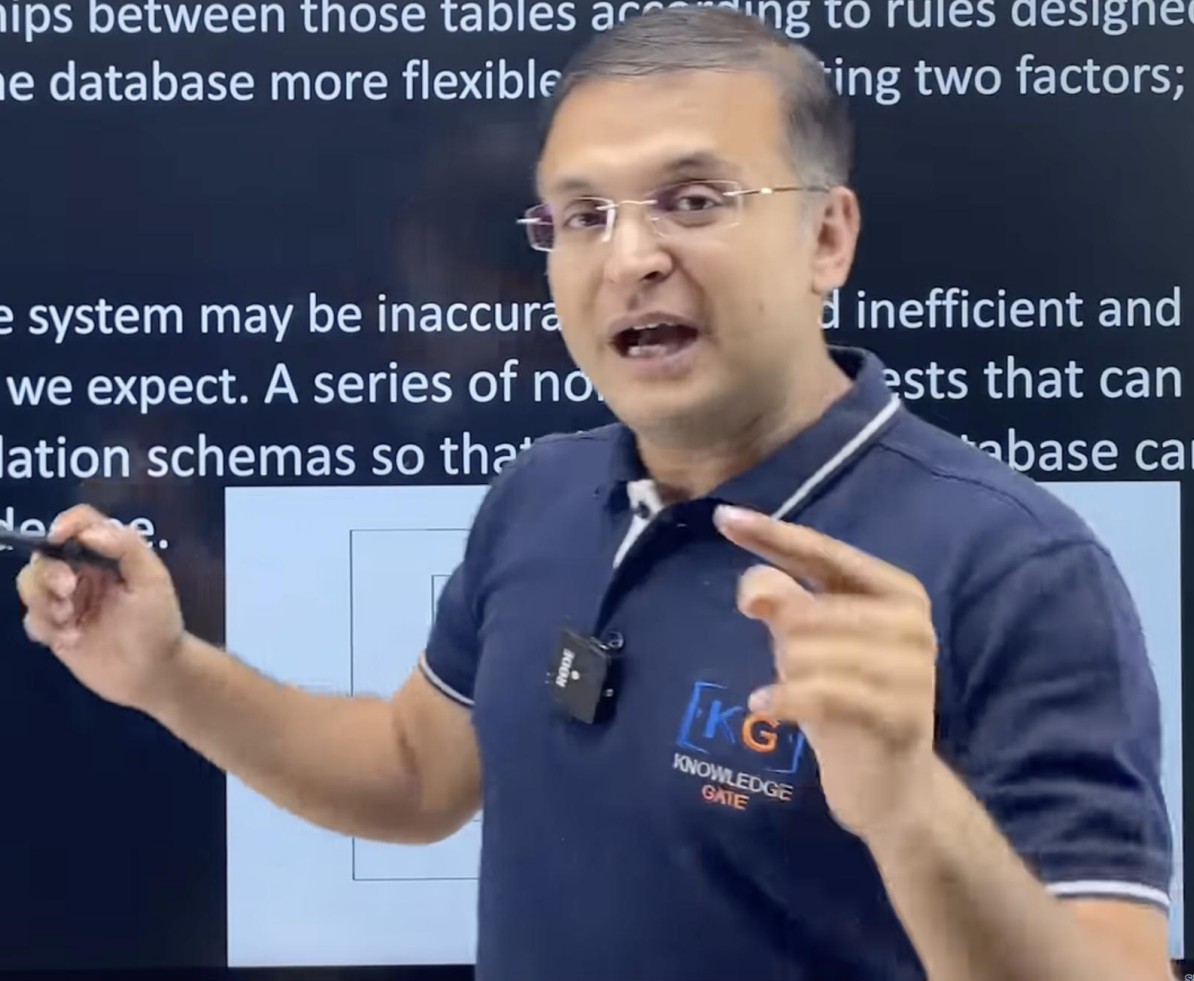
Roll no	name	Age	Br_code	Pr_name	Br_hod_name
1	A	19	101		Abc
2	B	18	101		Abc
3	C	20	101		Abc
4	D	20	102		Pqr

- Deletion Anomalies:** The deletion of a piece of information unintentionally removes other information.

Roll no	name	Age	Br_code	Br_name	Br_hod_name
1	A	19	101	Cs	Abc
2	B	18	101	Cs	Abc
3	C	20	101	Cs	Abc
4	D	20	102	Ec	Pqr

Purpose of Normalization

- Normalization may be simply defined as refinement process. Which includes creating tables and establishing relationships between those tables according to rules designed both to protect data and make the database more flexible. It is based on two factors;
 - Redundancy
 - Inconsistent Dependency
- With out normalization data base system may be inaccurate and inefficient and they might not produce the data we expect. A series of normalization tests that can be carried out on individual relation schemas so that the database can be **normalized** to any desired degree.
- 1NF>>>2NF>>3NF>>BCNF



Conclusion

- Like every paragraph must have a single idea similarly every table must have a single idea and if a table contains more than one idea then that table must be decomposed until each table contains only one idea. ✓

- We need some tools to approach this decomposition or normalization on large database which contains a number of table, and that tool is functional dependencies. ✓



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FUNCTIONAL DEPENDENCY

- A formal tool for analysis of relational schemas.
- In a Relation R, if ' α ' $\subseteq$ R AND ' β ' $\subseteq$ R, then
 - attribute or a Set of attribute ' α ' Functionally derives an attribute or set of attributes ' β ', iff each ' α ' value is associated with precisely one ' β ' value.
- For all pairs of tuples t_1 and t_2 in R such that
 - If $T_1[\alpha] = T_2[\alpha]$
 - Then, $T_1[\beta] = T_2[\beta]$

X	Y	Z
1	4	2
1	4	3
2	6	3
3	2	2

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Chapter-3 (Functional Dependency)

Functional Dependency

- Trivial Functional dependency - If β is a subset of α , then the functional dependency $\alpha \rightarrow \beta$ will always hold.

जिसका होना न होना बराबर हो

X	Y	Z
1		
1		
2		
3		



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ATTRIBUTES CLOSURE/CLOSURE ON ATTRIBUTE SET/ CLOSURE SET OF ATTRIBUTES

- Attribute closure of an attribute set can be defined as set of attributes which can be functionally determined from F.
 - DENOTED BY F^+

R (A, B, C, D)

$A \rightarrow B$

$B \rightarrow C$

$AB \rightarrow D$

$A^+ =$

R(ABCDEFGFG)

$A \rightarrow B$

$BC \rightarrow DE$

$AEG \rightarrow G$

$(AC)^+ = ?$

Q R(ABCDE)

$A \rightarrow BC$

$CD \rightarrow E$

$B \rightarrow D$

$E \rightarrow A$

$(B)^+ =$

ARMSTRONG'S AXIOMS

Armstrong's Axioms

1. An **axiom** or **postulate** is a statement that is taken to be true to serve as a premise or starting point for further reasoning and arguments.
2. **Armstrong's axioms** are a set of axioms (or, more precisely, inference rules) used to infer all the functional dependencies on a relational database. They were developed by William W. Armstrong in his 1974 paper.
3. The axioms are sound in generating only functional dependencies in the closure of a set of functional dependencies (denoted as F^+) when applied to that set (denoted as F).



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Armstrong Axioms

- **Reflexivity:** If Y is a subset of X , then $X \rightarrow Y$

Augmentation: If $X \rightarrow Y$, then $XZ \rightarrow YZ$

Transitivity: If $X \rightarrow Y$ and $Y \rightarrow Z$, then $X \rightarrow Z$

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From these rules, we can derive these secondary rules-

- **Union:** If $X \rightarrow Y$ and $X \rightarrow Z$, then $X \rightarrow YZ$
- **Decomposition:** If $X \rightarrow YZ$, then $X \rightarrow Y$ and $X \rightarrow Z$
- **Pseudo transitivity:** If $X \rightarrow Y$ and $WY \rightarrow Z$, then $WX \rightarrow Z$
- **Composition:** If $X \rightarrow Y$ and $Z \rightarrow W$, then $XZ \rightarrow YW$

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Chapter-3 (Functional Dependency)

Equivalence of Two FD

Q Consider the following set of fd R(ACDEH)

F	G
$A \rightarrow C$	$A \rightarrow CD$
$AC \rightarrow D$	
$E \rightarrow AD$	$E \rightarrow AH$
$E \rightarrow H$	

$$(A)^+ = ACD$$

$$(AC)^+ = AC\underline{D}$$

$$(E)^+ = E\underline{A}\underline{H}\underline{C}\underline{D}$$

$$(A)^+ = A\underline{C}\underline{D}$$

$$(E)^+ = E\underline{A}\underline{D}\underline{H}\underline{C}$$

$$F \subseteq G$$

$$G \subseteq F$$

$$F = G$$

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To find the MINIMAL COVER / CANONICAL COVER / IRREDUCIBLE SET

- A **canonical cover** (also known as a minimal cover) for a set of functional dependencies in a database is a minimal set of functional dependencies that is equivalent to the original set, but with redundant dependencies and extraneous attributes removed. It is used in the normalization process of database design to simplify the set of functional dependencies and to find a good set of relations.
- There may be any following type of redundancy in the set of functional dependencies: -
 - Complete production may be Redundant.
 - One or more than one attributes may be redundant on right hand side of a production.
 - One or more than one attributes may be redundant on Left hand side of a production.

Chapter-3 (Functional Dependency)

Canonical Cover

Q R(ABCD)

$A \rightarrow B$

$C \rightarrow B$

$D \rightarrow ABC$

$AC \rightarrow D$

$A \rightarrow B$ ✓

$A^+ = AB$
 $A^+ = A$

$C \rightarrow B$ ✓

$C^+ = CB$
 $C^+ = C$

$D \rightarrow A$

$D^+ = DABC$
 $D^+ = DBC$

~~$D \rightarrow B$~~ $D^+ = DACB$

~~$D \rightarrow C$~~ $D^+ = DAB$

$AC \rightarrow D$

$AC^+ = ACDB$

$A \rightarrow B$

$C \rightarrow B$

$D \rightarrow A$

$D \rightarrow C$

$AC \rightarrow D$

$(A)^+ = ACDB$

$(A)^+ = AB$

$(C)^+ = C$

$A \rightarrow B$

$C \rightarrow B$

$D \rightarrow AC$

$(A \rightarrow)$

Super key ✓

- Set of attributes using which we can identify each tuple uniquely is called Super key.
- Let X be a set of attributes in a Relation R , if X^+ (Closure of X) determines all attributes of R then X is said to be Super key of R .
- There should be at least one Super key in every relation.

$$R(ABCD)$$
$$AB \rightarrow C$$
$$(AB)^+ = ABC$$


A	B	C	D

Candidate key

- Minimal set of attributes using which we can identify each tuple uniquely is called Candidate key. A super key is called candidate key if it's No proper subset is super key. Also called as **MINIMAL SUPER KEY**.

should be at least one candidate key.

$R(A B C D)$

Prime attribute - Attributes that are member of at least one candidate Keys are called Prime attributes. ✓

FOR MORE CONTENT CHECKOUT www.youtube.com/channel/UCkGv...



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Primary key

- One of the candidate keys is selected by database administrator as a Primary means to identify tuple is called primary Key. Primary Key attribute are not allowed to have Null values. Exactly one Primary Key per table in RDMS.

Candidate key which are not chosen as primary key is alternate key.

Foreign Keys

- A foreign key is a column or group of columns in a relational database table that refers the primary key of the same table or some other table to represent relationship.



- The ***concept of referential integrity*** is derived from foreign key theory.



Chapter-3 (Functional Dependency)

- **Composite key** – Composite key is a key composed of more than one column sometimes it is also known as concatenated key.

- **Secondary key** – Secondary key is a key used to speed up the search and retrieval contrary to primary key, a secondary key does not necessarily contain unique values.



FIRST NORMAL FORM

- 1NF is the initial step of database normalization.
- Implications of first normal form
 - Atomic Values: Each cell in a table contains indivisible, atomic values. Means a Relation should not contain any multivalued or composite attributes.
 - Unique Columns: Each column must have a distinct name to identify the data it contains.
 - Primary Key: A table in 1NF should have a primary key that uniquely identifies each record.
 - Eliminating Duplicates: Duplicate rows are removed to prevent data redundancy.

(Chapter-4: Normalization)- 1NF, 2NF, 3NF, BCNF, Multivalued Dependency, 4NF, Lossy-Lossless Decomposition, 5NF, Dependency Preserving Decomposition.

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Chapter-4(Normalization)

PARTIAL DEPENDENCY- When a non – prime attribute is dependent only on a part (Proper subset) of candidate key then it is called partial dependency. (PRIME > NON-PRIME)

Full DEPENDENCY- When a non – prime attribute is dependent on the entire candidate key then it is called Full dependency.

①

e.g. R(ABCD)

AB → D

A → C

R(A B C D)

AB —

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(Chapter-4: Normalization)- 1NF, 2NF, 3NF, BCNF, Multivalued Dependency, 4NF, Lossy-Lossless Decomposition, 5NF, Dependency Preserving Decomposition.

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SECOND NORMAL FORM

- Relation R is in 2NF if,
 - R should be in 1 NF.
 - R should not contain any Partial dependency. (that is every non-prime attribute should be fully dependent upon candidate key)

View chapter

(Chapter-4: Normalization)- 1NF, 2NF, 3NF, BCNF, Multivalued Dependency, 4NF, Lossy-Lossless Decomposition, 5NF, Dependency Preserving Decomposition.

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TRANSITIVE DEPENDENCY – A functional dependency from non-Prime attribute to non-Prime attribute is called transitive

E.g.- R(A, B, C, D) with A as a candidate key

$A \rightarrow B$

$B \rightarrow C$ [transitive dependency]

$C \rightarrow D$ [transitive dependency]

THIRD NORMAL FORM

- Let R be the relational schema, it is said to be in 3 NF
 - R should be in 2NF
 - It must not contain any transitive dependency

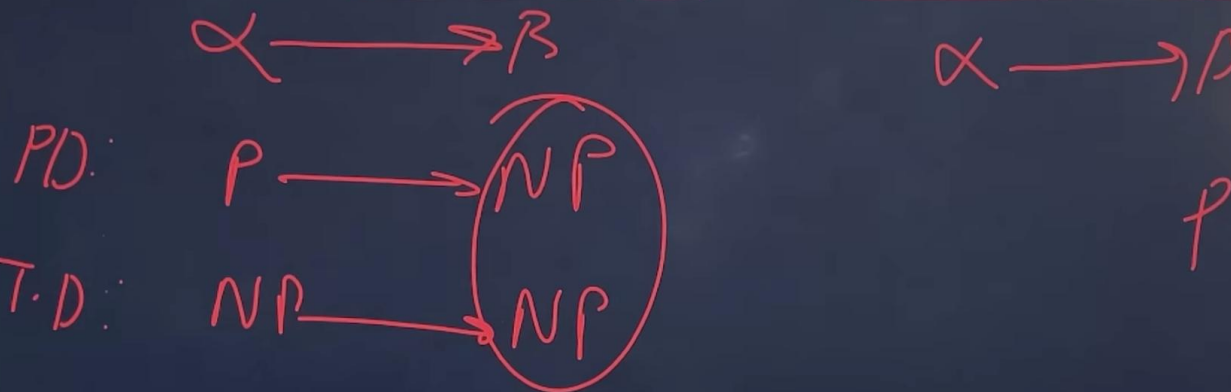


THIRD NORMAL FORM DIRECT DEFINATION

- A relational schema R is said to be 3 NF if every functional dependency in R from $\alpha \rightarrow \beta$, either α is super key or β is the prime attribute

THIRD NORMAL FORM DIRECT DEFINATION

- A relational schema R is said to be 3 NF if every functional dependency in R from $\alpha \rightarrow \beta$, either α is super key or β is the prime attribute



BCNF (BOYCE CODD NORMAL FORM)

- A relational schema R is said to be BCNF if every functional dependency in R from
 - $\alpha \rightarrow \beta$
 - α must be a super key



Some important note points on Normalization:

- A Relation with two attributes is always in BCNF. ✓
- A Relation schema R consist of only prime attributes then R is always in 3NF, but may or may not be in BCNF.